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DUNMAN HIGH SCHOOL Preliminary Examination Year 6

H1 CHEMISTRY

8872/01

Paper 1 Multiple Choice

27 September 2011

50 minutes

Additional Materials: Optical Mark Sheet
 Data Booklet

INSTRUCTIONS TO CANDIDATES

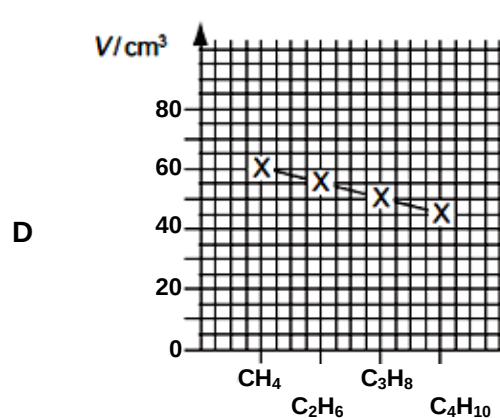
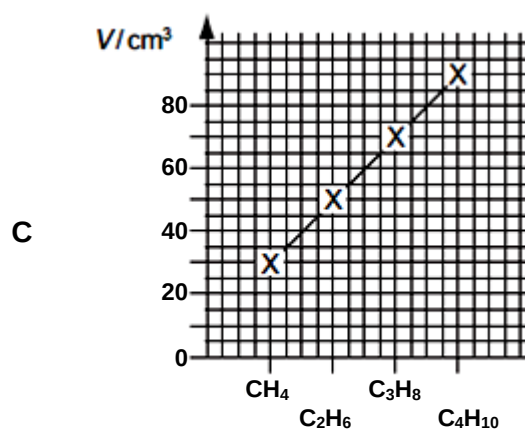
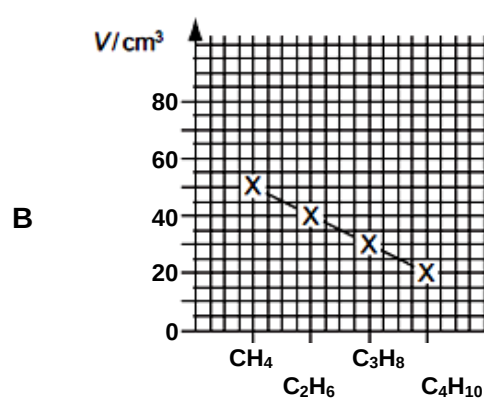
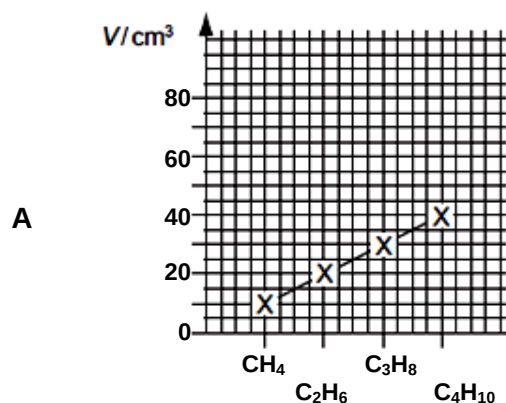
- 1 Write your **name** and **class** on this question paper.
- 2 There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
- 3 Each correct answer will score one mark. A mark will not be deducted for wrong answer.
- 4 Any rough working should be done in this booklet.
- 5 You may use a calculator.

Section A

For each question, there are four possible answers **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 Samples of 10 cm^3 of each of the first four members of the alkane series are separately mixed with 70 cm^3 of oxygen. Each is then burned and the total volume, V , of the residual gas measured again at room temperature and pressure.

Which graph represents the results that would be obtained?



- 2 If $\frac{1}{10}$ mol of ethanol, $\frac{1}{8}$ mol of hydrogen bromide are used to prepare bromoethane, what is the maximum mass of bromoethane which could be formed?
[M_r of ethanol: 46; M_r of bromoethane: 109]

A $\frac{109}{10} \text{ g}$

B $\frac{109}{8} \text{ g}$

C $\frac{109}{10 \times 46} \text{ g}$

D $\frac{109}{8 \times 46} \text{ g}$

- 3 The sea worm *Ascidia nigra* accumulates vanadium from seawater to an enormous extent. The concentration of vanadium in its blood cell is 14 g dm^{-3} and each animal has about 3 ml of blood. One litre of seawater contains $7.1 \text{ } \mu\text{g}$ of vanadium.
[$1 \text{ } \mu\text{g} = 10^{-6} \text{ g}$]

Assuming that *Ascidia nigra* extract 100% of the vanadium, how many litres of sea water does each animal have to process to achieve this concentration in their blood?

- A 0.0059 l B 5.915 l C 591.5 l D 5915 l

- 4 In an historically famous experiment Wöhler heated 'inorganic' ammonium cyanate in the absence of air. The only product of the reaction was 'organic' urea, $\text{CO}(\text{NH}_2)_2$. No other products were formed in the reaction.

What is the formula of the cyanate ion present in ammonium cyanate?

- A CNO^-
B CNO^{2-}
C CO^-
D NO^-

- 5 In some fireworks there is a reaction between powdered aluminium and powdered barium nitrate in which heat is evolved to result in decomposition of both nitrates and production of an unreactive gas.

What is the equation for this reaction?

- A $2\text{Al} + \text{Ba}(\text{NO}_3)_2 \rightarrow \text{Al}_2\text{O}_3 + \text{BaO} + 2\text{NO}$
B $4\text{Al} + 4\text{Ba}(\text{NO}_3)_2 \rightarrow 2\text{Al}_2\text{O}_3 + 4\text{Ba}(\text{NO}_2)_2 + \text{O}_2$
C $10\text{Al} + 3\text{Ba}(\text{NO}_3)_2 \rightarrow 5\text{Al}_2\text{O}_3 + 3\text{BaO} + 3\text{N}_2$
D $10\text{Al} + 18\text{Ba}(\text{NO}_3)_2 \rightarrow 10\text{Al}(\text{NO}_3)_3 + 18\text{BaO} + 3\text{N}_2$

- 6 Use of the *Data Booklet* is relevant to this question.

When aqueous hydrogen peroxide, H_2O_2 , is mixed with acidified potassium dichromate(VI), there is a colour change from orange to green. When aqueous hydrogen peroxide is added to acidified potassium iodide solution, there is a colour change from colourless to brown.

The oxidation number of oxygen in H_2O_2 is -1 ,

What are the oxidation numbers of oxygen after the reactions with potassium dichromate(VI) and potassium iodide?

	after reaction with potassium dichromate(VI)	After reaction with potassium iodide
A	-2	-2
B	-2	0
C	0	-2
D	0	0

- 7 The use of *Data Booklet* is relevant to this question.

The successive ionisation energies, in kJ mol^{-1} , of an element X are given below.

870 1800 3000 3600 5800 7000 13200

What is X?

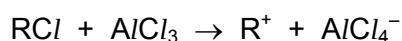
A ${}_{33}\text{As}$

B ${}_{53}\text{I}$

C ${}_8\text{O}$

D ${}_{52}\text{Te}$

- 8 Aluminium chloride catalyses certain reactions by forming carbocations (carbonium ions) with chloroalkanes as shown.



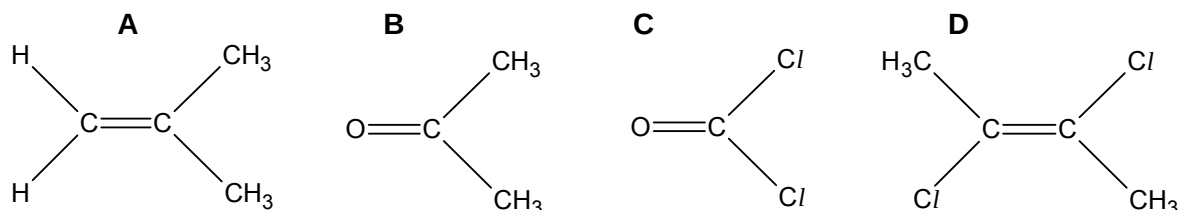
Which property makes this reaction possible?

- A** AlCl_3 exists as the dimer Al_2Cl_6 in the vapour.
- B** AlCl_3 is a covalent molecule.
- C** The aluminium atom in AlCl_3 has an incomplete octet of electrons.
- D** The chlorine atom in RCl has a vacant p orbital.

9 Which one of the following structural features is common to both diamond and graphite?

- A Covalent bonds between carbon atoms.
- B Delocalised electrons.
- C Each carbon atom bonded to four others.
- D van der Waals' forces.

10 Which molecule has the largest dipole?



11 Why does copper wire conduct electricity when a potential difference is applied to it?

- A Atoms of copper become ionised.
- B Bonding electrons in the crystal lattice move.
- C Copper(II) ions move to the cathode.
- D The crystal lattice breaks down.

12 Which of the following does **not** involve the lone pair of electrons on the nitrogen atom of the ammonia molecule?

- A The formation of NH_4^+ ions in aqueous ammonia.
- B The formation of an addition product between boron trifluoride and ammonia.
- C The formation of NH_2^- ions during the reaction of ammonia with sodium.
- D The hydrogen bonding that occurs between molecules of ammonia.

13 Which of the following has the same value as the standard enthalpy change of formation of carbon monoxide?

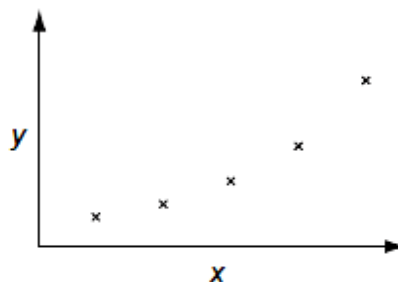
[ΔH_c^θ and ΔH_f^θ represent the standard enthalpy change of combustion and formation respectively.]

- A $\Delta H_c^\theta (\text{C(graphite)}) - \Delta H_c^\theta (\text{CO})$
- B $\Delta H_f^\theta (\text{CO}_2) - \Delta H_c^\theta (\text{C(graphite)})$
- C $\Delta H_f^\theta (\text{CO}_2) - \frac{1}{2}\Delta H_c^\theta (\text{C(graphite)})$
- D $\frac{1}{2}\Delta H_c^\theta (\text{C(graphite)})$

- 14 The reaction between **P** and **Q** in aqueous solution is acid-catalysed, and the reaction is first order with respect to the hydrogen ion. If all other conditions are kept constant, what is the ratio of

$$\frac{\text{rate of reaction at pH 3}}{\text{rate of reaction at pH 1}} ?$$

- A 0.01 B 0.33 C 1 D 3
- 15 Which group of particles is in order of increasing size?
- A N O F
- B N^{3-} O^{2-} F^-
- C Na^+ Mg^{2+} Al^{3+}
- D Na^+ Ne F^-
- 16 Use of the *Data Booklet* is relevant to this question.
The sketch graph shows the variation of one physical or chemical property with another for the Period 3 elements.



What are the correct labels for the axes?

	x-axis	y-axis
A	atomic number	mass number
B	atomic number	melting point
C	first ionisation energy	atomic number
D	first ionisation energy	atomic radius

17 A compound **U** has all of the properties below.

- It is a liquid at 25 °C.
- It mixes completely with water.
- It reacts with aqueous sodium hydroxide.

What could **U** be?

- A** ethanoic acid
B ethanol
C ethanal
D ethyl ethanoate

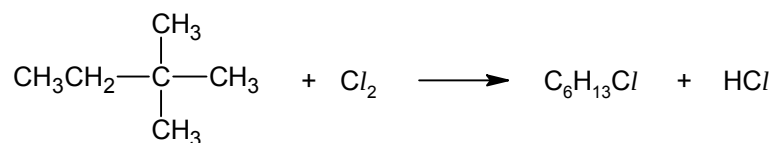
18 In which class of compound, in its general formula, is the ratio of hydrogen atoms to carbon atoms the highest?

- A** alcohols
B aldehydes
C carboxylic acids
D halogenoalkanes

19 Which property does benzene have as a consequence of the delocalised electrons present in its molecule?

- A** Addition reactions of benzene take place more easily than substitution.
B Benzene is a good conductor of electricity.
C Substitution in benzene takes place at one particular carbon atom.
D The carbon–carbon bond lengths are between those of C–C bonds and C=C bonds.

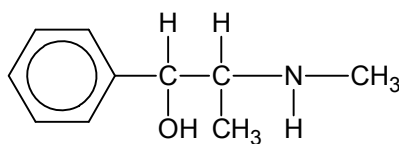
20 When heated with chlorine, the hydrocarbon 2,2–dimethylbutane undergoes free radical substitution.



How many different monochloro–substituted products can be obtained?

- A** 1 **B** 2 **C** 3 **D** 4

- 21 Ephedrine is a drug that is weakly basic and widely used in cold and allergy medications.

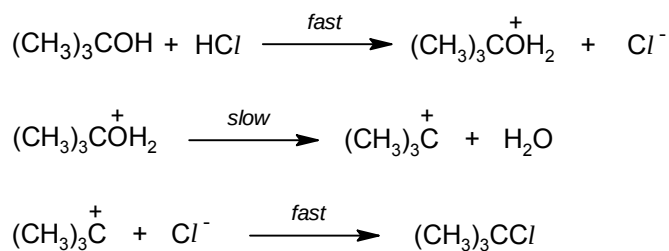


Which of the following pairs of reagents consists of one which reacts with ephedrine and one which does not react with ephedrine?

- | | | |
|----------|--------------------------------|--------------------------|
| A | CH_3COOH | CH_3COCl |
| B | CH_3Br | NaOH (aq) |
| C | HCl(aq) | PCl_5 |
| D | $[\text{Ag}(\text{NH}_3)_2]^+$ | Br_2 |
- 22 **P**, **Q** and **R** are three different 3-carbon compounds. **P** and **Q** react together to form an ester. **R** reacts with **P** to give the same ester but much *less* readily when compared to **Q**.

Compound **Q** could be

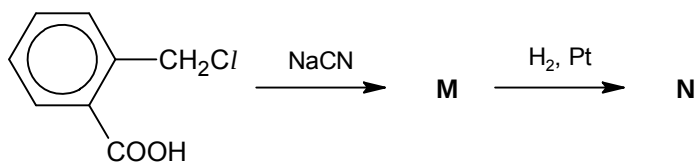
- | | | | |
|----------|----------------|----------|-------------------|
| A | propanoic acid | B | propanone |
| C | propan-2-ol | D | propanoyl bromide |
- 23 The mechanism for a certain reaction is given below.



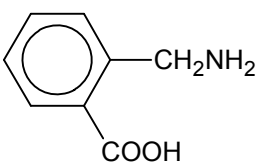
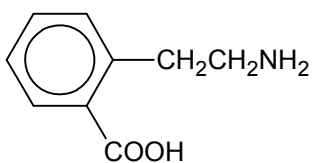
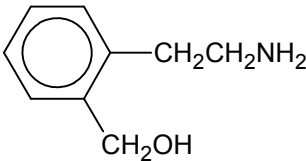
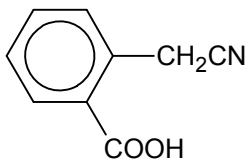
Which of the following statement is **not** true?

- A** $(\text{CH}_3)_3\text{COH}_2^+$ and $(\text{CH}_3)_3\text{COH}$ are conjugate acid-base pair.
- B** HCl donates an electron pair to alcohol.
- C** The overall reaction is $(\text{CH}_3)_3\text{COH} + \text{HCl} \rightarrow (\text{CH}_3)_3\text{CCl} + \text{H}_2\text{O}$.
- D** The above mechanism is a nucleophilic substitution reaction.

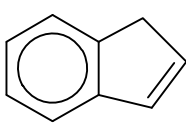
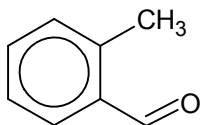
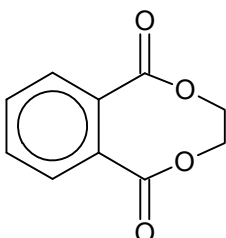
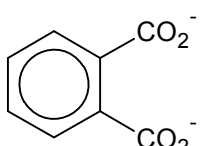
- 24 A reaction sequence is shown below.



What will be the product **N**?

- A** 
B 
- C** 
D 

- 25 Which of the following organic compound will **not** produce 1,2-benzene-dicarboxylic acid when subjected to the reagents and conditions as specified?

- A**  , KMnO_4/H^+ , heat
- B**  , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$, heat
- C**  , H^+ (aq), heat
- D**  , H^+ (aq)

Section B

For each question, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

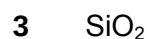
The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

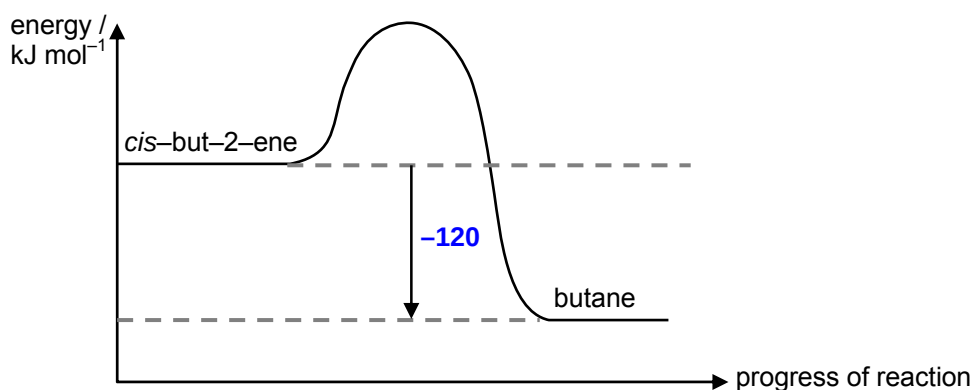
No other combination of statements is used as a correct response.

- 26 In microwave ovens, the wave energy is absorbed by certain polar molecules.

Which of the following would absorb microwave energy?



- 27 The enthalpy change of hydrogenation, $\Delta H_{\text{hydrogenation}}^\theta$ of *cis*-but-2-ene is found to be -120 kJ mol^{-1} . When a mixture of *cis*- and *trans*-but-2-ene is allowed to undergo hydrogenation to form butane, the overall $\Delta H_{\text{hydrogenation}}^\theta$ is found to be -117 kJ mol^{-1} . It is known that *cis*-but-2-ene is less stable than the *trans*-isomer by 4 kJ mol^{-1} .



Which of the following statement is correct?

- 1 $\Delta H_{\text{hydrogenation}}^\theta$ of *trans*-but-2-ene is -120 kJ mol^{-1} .
- 2 $\Delta H_{\text{hydrogenation}}^\theta$ value can be used to determine stability of *cis*-*trans* isomers.
- 3 The fraction of *cis* but-2-ene in the mixture is 0.25.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 Given the following bond energies:

<i>bond</i>	<i>bond energy / kJ mol⁻¹</i>
C–C	350
C=C	610

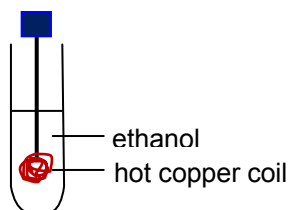
What can be deduced from the data?

- 1** The π bond is stronger than the σ bond.
- 2** The π bond energy can be determined from the data.
- 3** The $\text{C}\equiv\text{C}$ bond energy is estimated to be 870 kJ mol^{-1} .

29 2-methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$, is more difficult to oxidise than propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$. This is because 2-methylpropan-2-ol

- 1** does not contain the >CH—OH grouping.
- 2** has more methyl groups, which are much larger than hydrogen atoms, hence posing steric hindrance to the attacking oxidising agent.
- 3** has a higher boiling point.

30 A piece of hot copper coil is repeatedly plunged into ethanol as shown in the diagram below. To the resulting colourless solution **L**, ammonical silver nitrate is added and a silver mirror is observed. Which of the following statement is correct?



- 1** Both ethanol and solution **L** when reacted with alkaline aqueous iodine will give a yellow solid.
- 2** Ethanol is oxidised on the surface of the copper coil.
- 3** The copper is reduced in the above reaction.

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DHS H1 CHEMISTRY YEAR 6 PRELIMINARY EXAMINATION 2011

Paper 1 MCQ – Answers

1	2	3	4	5	6	7	8	9	10
D	A	D	A	C	C	D	C	A	B

11	12	13	14	15	16	17	18	19	20
B	C	A	A	D	A	A	A	D	C

21	22	23	24	25	26	27	28	29	30
B	D	B	B	B	D	C	C	D	B